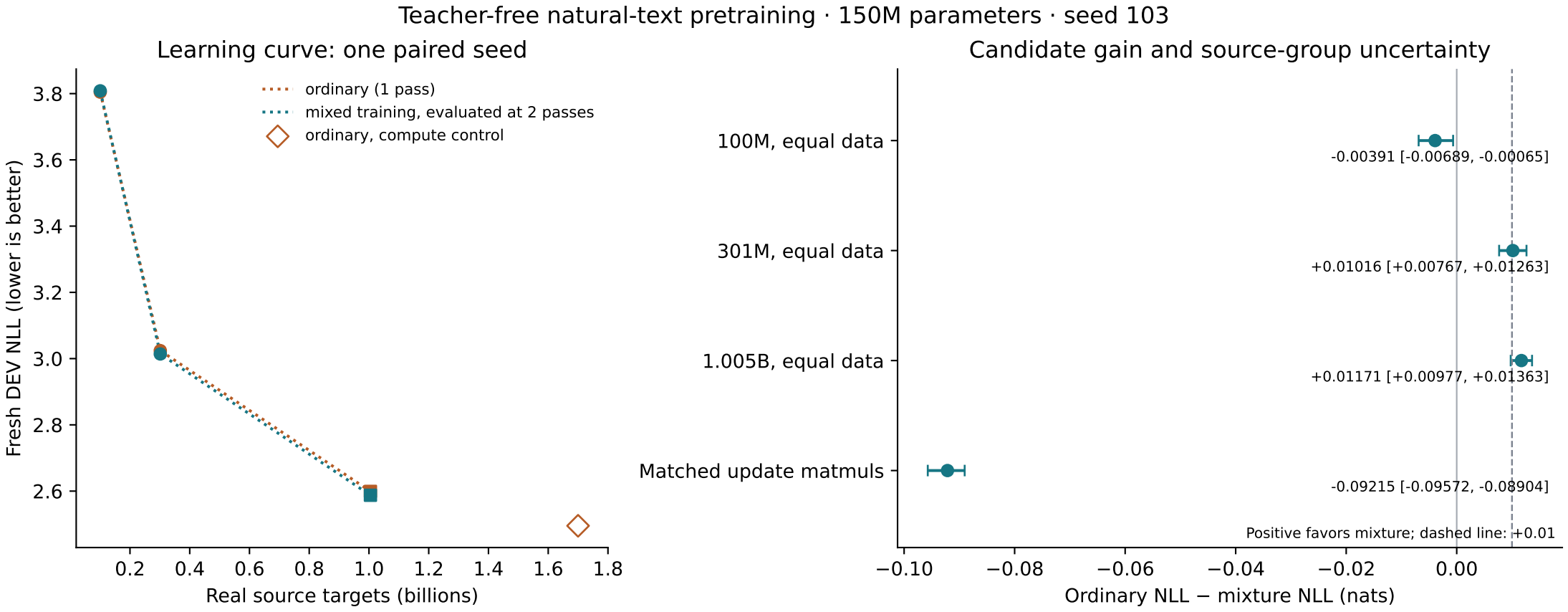
Teacher-free natural-text pretraining · 150M parameters · seed 103
Learning curve: one paired seed
ordinary (1 pass)
mixed training, evaluated at 2 passes
ordinary, compute control
Fresh DEV NLL (lower is better)
3.8
3.6
3.4
3.2
3.0
2.8
2.6
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Real source targets (billions)
Candidate gain and source-group uncertainty
100M, equal data
-0.00391 [-0.00689, -0.00065]
301M, equal data
+0.01016 [+0.00767, +0.01263]
1.005B, equal data
+0.01171 [+0.00977, +0.01363]
Matched update matmuls
-0.09215 [-0.09572, -0.08904]
Positive favors mixture; dashed line: +0.01
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
Ordinary NLL − mixture NLL (nats)
Circles: stable LR. Squares/diamond: completed decay. The 1.7B control branches from 904M stable, not the 1B endpoint.
Intervals: source-group bootstrap conditional on fitted weights; training-seed uncertainty remains pending.